Assignment 4 — Security, Resilience, and Intrusion Tolerance

Synthetic Kazakhstan University Research Server Incident

Astana IT University | Fault Tolerance | Academic Year 2026–2027

Context note: All operational data in this assignment are synthetic educational data. They are designed to represent a realistic Kazakhstan/Astana university context and do not describe actual AITU infrastructure, incidents, or internal systems.

Scenario

A synthetic research server used by a Kazakhstan university research group has experienced a ransomware-style incident. The task is to analyze the incident, design resilient security controls, and evaluate recovery.

The dataset is fictional and is not an account of an actual Astana IT University or Kazakhstan university cyber incident.

Incident dataset

Time	Event	Source	Severity
09:01	15 failed logins	10.20.4.15	Medium
09:03	21 failed logins	10.20.4.15	High
09:04	Successful login	10.20.4.15	High
09:07	320 files modified	10.20.4.15	Critical
09:08	CPU usage increased to 95%	Research-Server-01	High
09:10	1,200 files encrypted	Research-Server-01	Critical
09:12	Backup deletion attempt	Research-Server-01	Critical
09:13	Outbound connection to unknown IP	Research-Server-01	High
09:15	Database access spike	Research-Server-01	High
09:18	Antivirus disabled	Research-Server-01	Critical

Part A — Incident Analysis (6 points)

Classify each event as reconnaissance, credential attack, privilege compromise, lateral movement, data destruction, persistence, exfiltration, or recovery attack.

Time	Event	Attack stage	Evidence	Risk
09:01				
09:03				
09:04				
09:07				
09:08				
09:10				
09:12				
09:13				
09:15				
09:18				

Part B — Risk Matrix (6 points)

Risk = Likelihood × Impact. Use 1–4 Low, 5–9 Medium, 10–16 High, 17–25 Critical.

Threat	Likelihood	Impact	Risk Score	Priority
Credential compromise				
Ransomware				
Backup destruction				
Data exfiltration				
Privilege escalation				

Part C — Resilient Security Architecture (7 points)

Design a synthetic architecture containing MFA, network segmentation, immutable backup, monitoring, intrusion detection, secure failover, and a recovery environment.

Threat	Preventive control	Detection control	Recovery control
Credential attack			
Ransomware			
Backup deletion			
Data exfiltration			
Privilege escalation			

Part D — Recovery Plan (6 points)

Requirements: RPO ≤ 5 minutes; RTO ≤ 30 minutes.

Step	Action	Responsible component	Target time
1	Detect incident		
2	Isolate affected system		
3	Preserve evidence		
4	Activate backup		
5	Restore services		
6	Verify integrity		
7	Return to production		

Part E — AI-Assisted Detection (5 points)

Event	Normal score	Observed score
Login failures	0.10	0.81
File modifications	0.12	0.96
CPU usage	0.20	0.88
Outbound traffic	0.15	0.79
Database requests	0.18	0.91

Assume score ≥ 0.75 indicates an anomaly.

1. Calculate the number and rate of detected anomalies.
2. Identify the highest-priority event.
3. Explain why AI-assisted detection should complement, not replace, deterministic security and recovery controls.
4. Recommend one additional resilience mechanism appropriate for a Kazakhstan university research environment.

Submission

- Completed incident-analysis and risk tables
- Security architecture
- Recovery plan
- AI anomaly analysis
- Short conclusion
- Oral defense

STUDENT TECHNICAL WORK REPORT / SUBMISSION RECORD

Assignment 4 — Security, Resilience, and Intrusion Tolerance

Purpose: This form is a formal submission record. The student must complete every applicable field and attach evidence of their own work. Short generic explanations are not sufficient.

1. Student and Submission Information

Required field	Student entry
Student full name	
Student ID	
Group / cohort	
Instructor	
Date of submission	____ / ____ / _____
Individual variant / task code assigned by instructor	
Git repository / project link	
Attached code / file names	

2. Work Period and Task Record

Item	Student entry
Work start date/time	
Work completion date/time	
Main task completed	
Tools / technologies used	
Main technical output produced	

3. Report Structure to Complete

For each section below, write the requested evidence in your own words. Include calculations, diagrams, code references, screenshots, logs, or experimental evidence where applicable.

1. Incident Analysis

Classify every incident event into an attack stage, provide evidence from the synthetic dataset, and assess risk.

Required content	Student response / evidence
Summary	
Evidence / calculation / artifact reference	
Student-specific decision or observation	

2. Risk Matrix

Assign likelihood and impact, calculate Risk = Likelihood × Impact, and justify the resulting priority.

Required content	Student response / evidence

Summary	
Evidence / calculation / artifact reference	
Student-specific decision or observation	

3. Resilient Security Architecture

Design the required architecture containing MFA, segmentation, immutable backup, monitoring, intrusion detection, secure failover, and recovery environment.

Required content	Student response / evidence
Summary	
Evidence / calculation / artifact reference	
Student-specific decision or observation	

4. Recovery Plan

Complete the seven-step recovery plan and show how your target RPO ≤ 5 minutes and RTO ≤ 30 minutes are addressed.

Required content	Student response / evidence
Summary	
Evidence / calculation / artifact reference	
Student-specific decision or observation	

5. AI-Assisted Detection

Calculate the anomaly count/rate, identify the highest-priority event, and explain the role and limitations of AI-assisted detection.

Required content	Student response / evidence
Summary	

Evidence / calculation / artifact reference	
Student-specific decision or observation	

6. Additional Resilience Recommendation

Recommend one additional resilience mechanism appropriate for the synthetic Kazakhstan university research environment and justify it.

Required content	Student response / evidence
Summary	
Evidence / calculation / artifact reference	
Student-specific decision or observation	

4. Individual Work and Anti-Plagiarism Record

Because the assignment uses a common educational scenario and dataset, the submitted report must contain verifiable evidence of the individual student's work. Copying another student's wording, calculations, architecture, code, logs, screenshots, or conclusions is not acceptable.

- Use your own threat/control mapping rather than copying another student's table wording.
- Tie each risk score to your own justification.
- Provide a unique architecture diagram and explain its control flow.
- Record the exact source file/version used for your analysis.
- Be prepared to defend why one control is preventive, detective, or recovery-oriented.

Individual evidence	Student entry
My unique technical decision	
My own implementation / calculation evidence	
Exact file, code, commit, notebook, or log reference	
One problem I personally encountered and how I solved it	
One limitation of my own solution	

5. Evidence Checklist

- ☐ Completed required tables and calculations
- ☐ Architecture / RBD / FTA / security diagram where applicable
- ☐ Source code or configuration files where applicable
- ☐ Execution logs / experiment results where applicable

- ☐ Screenshots or other reproducible evidence where applicable
- ☐ Links to code, datasets, notebooks, or artifacts
- ☐ Before/after comparison where applicable
- ☐ Final technical conclusion

6. Student Declaration

I confirm that this report describes my own work. I have not copied another student's report, code, calculations, diagrams, logs, screenshots, or conclusions. Any external source or reused material has been identified. I understand that I may be asked to reproduce or explain any part of this submission during the oral defense.

Role	Name	Signature	Date
Student / Executor	_____	_____	_____
Instructor / Accepted by	_____	_____	_____

7. Instructor Acceptance Record

Check	Status / comment
Report structure completed	☐ Yes ☐ No Comment: _____
Evidence is sufficient	☐ Yes ☐ No Comment: _____
Individual contribution is identifiable	☐ Yes ☐ No Comment: _____
Oral defense completed	☐ Yes ☐ No Comment: _____
Final status	☐ Accepted ☐ Revision required